

Esercizi capitolo 19

$$N_A = 6,02 \cdot 10^{23} \text{ mol}^{-1}$$

$$M = m N_A$$

$$n = \frac{N}{N_A} = \frac{M_{\text{campione}}}{M} = \frac{M_{\text{campione}}}{m N_A}$$

1) $N = 7,5 \cdot 10^{24}$

$$M = 74,9 \text{ g/mol}$$

$$m = \frac{M}{N_A} \cdot N = \frac{74,9 \text{ g/mol}}{6,02 \cdot 10^{23}} \cdot 7,5 \cdot 10^{24} = 0,93 \text{ kg}$$

2) $M = 197 \text{ g/mol}$ / m /

$$m = 2,50 \text{ g}$$

N /

$$n = \frac{2,50 \text{ g}}{197 \text{ g/mol}} = 0,0127 \text{ mol}$$

$$N = n N_A = 0,0127 \text{ mol} \cdot 6,02 \cdot 10^{23} \text{ mol}^{-1} = 7,64 \cdot 10^{21}$$

Gas perfetti:

$$pV = nRT$$

oppure

$$pV = NkT$$

T in Kelvin

$$R = 8,31 \text{ J/(mol} \cdot \text{K)}$$

$$k = 1,38 \cdot 10^{-23} \text{ J/K}$$

$$3) V_i = 1000 \text{ cm}^3 = 0.001 \text{ m}^3$$

$$T_i = 40,0^\circ\text{C} = 313,15^\circ\text{K}$$

$$p_i = 1,01 \cdot 10^5 \text{ Pa}$$

$$V_f = 1500 \text{ cm}^3 = 0.0015 \text{ m}^3$$

$$p_f = 1,06 \cdot 10^5 \text{ Pa}$$

m?

T_f ?

Risolto:

$$pV = nRT \Rightarrow n = \frac{pV}{RT} = \frac{1,01 \cdot 10^5 \cdot 0.001}{8,31 \cdot 313,15} = 0,04 \text{ mol}$$

$$p_f V_f = nRT_f \Rightarrow T_f = \frac{p_f V_f}{nR} = \frac{1,06 \cdot 10^5 \cdot 0,0015}{0,04 \cdot 8,31} = 478,3^\circ\text{K}$$

$$4) T = 10^\circ\text{C} = 283,15^\circ\text{K}$$

$$p = 100 \text{ kPa} = 1 \cdot 10^5 \text{ Pa}$$

$$V = 2,5 \text{ m}^3$$

n?

$$\text{Se } p_f = 3 \cdot 10^5 \text{ Pa e}$$

$$T_f = 303,15^\circ\text{K}, V_f?$$

$$pV = nRT \Rightarrow n = \frac{pV}{RT} = \frac{1 \cdot 10^5 \cdot 2,5}{8,31 \cdot 283,15} = 106,25 \text{ mol}$$

$$p_f V_f = nRT_f \Rightarrow V_f = \frac{nRT_f}{p_f} = \frac{106,25 \cdot 8,31 \cdot 303,15}{3 \cdot 10^5} =$$

$$= 0,89 \text{ m}^3$$

$$5) p = 1,01 \cdot 10^{-13} \text{ Pa}$$

$$T = 293 \text{ K}$$

$$V = 1 \text{ cm}^3 = 1 \cdot 10^{-6} \text{ m}^3$$

$N?$

$$pV = NkT \Rightarrow N = \frac{pV}{kT} = \frac{1,01 \cdot 10^{-13} \cdot 10^{-6}}{1,38 \cdot 10^{-23} \cdot 293} = 24,98$$

$$6) T = 5^\circ \text{C} = 278,15^\circ \text{K}$$

$$T_{\text{auto}} = 75^\circ \text{C} = 348,15^\circ \text{K}$$

$$pV = nRT$$

$$V_f = V_i \Rightarrow \frac{p_f}{p_i} = \frac{nRT_f}{nRT_i}$$

$$\Rightarrow p_f = \frac{T_f}{T_i} \cdot p_i = \frac{348,15^\circ \text{K}}{278,15^\circ \text{K}} \cdot 10^5 \text{ Pa} = 1,25 \cdot 10^5 \text{ Pa}$$

$$7) n = 1,8 \text{ mol}$$

$$V_i = 3 \text{ m}^3$$

$$V_f = 1,5 \text{ m}^3$$

$$T = 30^\circ \text{C} = 303,15^\circ \text{K}$$

Compressione isoterma

$Q?$

$$\Delta E_{int} = Q - L$$

$$\Delta E_{int} = 0 \Rightarrow Q = L$$

$$L = n R T \ln\left(\frac{V_f}{V_i}\right) = 1,2 \cdot 8,31 \cdot 30 \cdot \ln\left(\frac{1,3}{3}\right) = -311 \text{ J} = Q$$

Trasferimento del gas.

$$8) V = 1 \text{ cm}^3 = 1 \cdot 10^{-6} \text{ m}^3$$

$$p = 100 \text{ Pa}$$

$$T = 220^\circ \text{ K}$$

$$p V = n R T \Rightarrow n = \frac{p V}{R T} = \frac{100 \cdot 1 \cdot 10^{-6}}{8,31 \cdot 220} = 5,4 \cdot 10^{-8} \text{ mol}$$

$$n = \frac{N}{N_A} \Rightarrow N = n N_A = 5,4 \cdot 10^{-8} \cdot 6,02 \cdot 10^{23} = 3,25 \cdot 10^{16}$$

$$9) V = 1,64 \cdot 10^{-2} \text{ m}^3$$

$$p = 165 \text{ kPa} = 165 \cdot 10^3 \text{ Pa}$$

$$T = 0^\circ \text{ C} = 273,15^\circ \text{ K}$$

$$T_2 = 270^\circ \text{ C} = 543,15^\circ \text{ K}$$

$$V_f = 1,67 \cdot 10^{-2} \text{ m}^3$$

$$p = 165 \cdot 10^3 \text{ Pa} = 165 \cdot 10^3 \text{ Pa}$$

pe?

risolvo: $pV = nRT$

$$p_i = p + p_{et} = 165 \cdot 10^3 \text{ Pa} + 101 \cdot 10^3 \text{ Pa} = 266 \cdot 10^3 \text{ Pa}$$

$$n = \frac{pV}{RT} = \frac{266 \cdot 10^3 \cdot 1,64 \cdot 10^{-2}}{8,31 \cdot 273,15} = 1,92 \text{ mol}$$

$$p_f = \frac{nRT_f}{V_f} - p_{et} = \frac{1,92 \cdot 8,31 \cdot 543,13}{1,67 \cdot 10^{-2}} - 101 \cdot 10^3 =$$

$$= 417\,926 \text{ Pa}$$

11) $V_i = 0,140 \text{ m}^3$

$$p_{rel} = 103 \text{ kPa} = 103 \cdot 10^3 \text{ Pa}$$

$$p_{et,f} = 101,3 \text{ kPa} = 101,3 \cdot 10^3 \text{ Pa}$$

$L?$

ricordiamo $pV = nRT$ $\Delta E_{int} = Q - L$

$$p_{et,i} = p_{rel} + 101 \cdot 10^3 \text{ Pa} = 103 \cdot 10^3 \text{ Pa} + 101 \cdot 10^3 \text{ Pa} = 204 \cdot 10^3 \text{ Pa}$$

$$L_{tot} = L_{isoterma} + L_{isobara}$$

$$\begin{cases} V_i = \frac{nRT}{p_i} \\ V_f = \frac{nRT}{p_f} \end{cases}$$

$$\frac{V_f}{V_i} = \frac{p_i}{p_f}$$

Isoterme:

$$L = p_i V_i \ln\left(\frac{p_i}{p_f}\right) = (204 \cdot 10^3)(0,140) \ln\left(\frac{204 \cdot 10^3}{101,3 \cdot 10^3}\right) = 199935$$

Isobare:

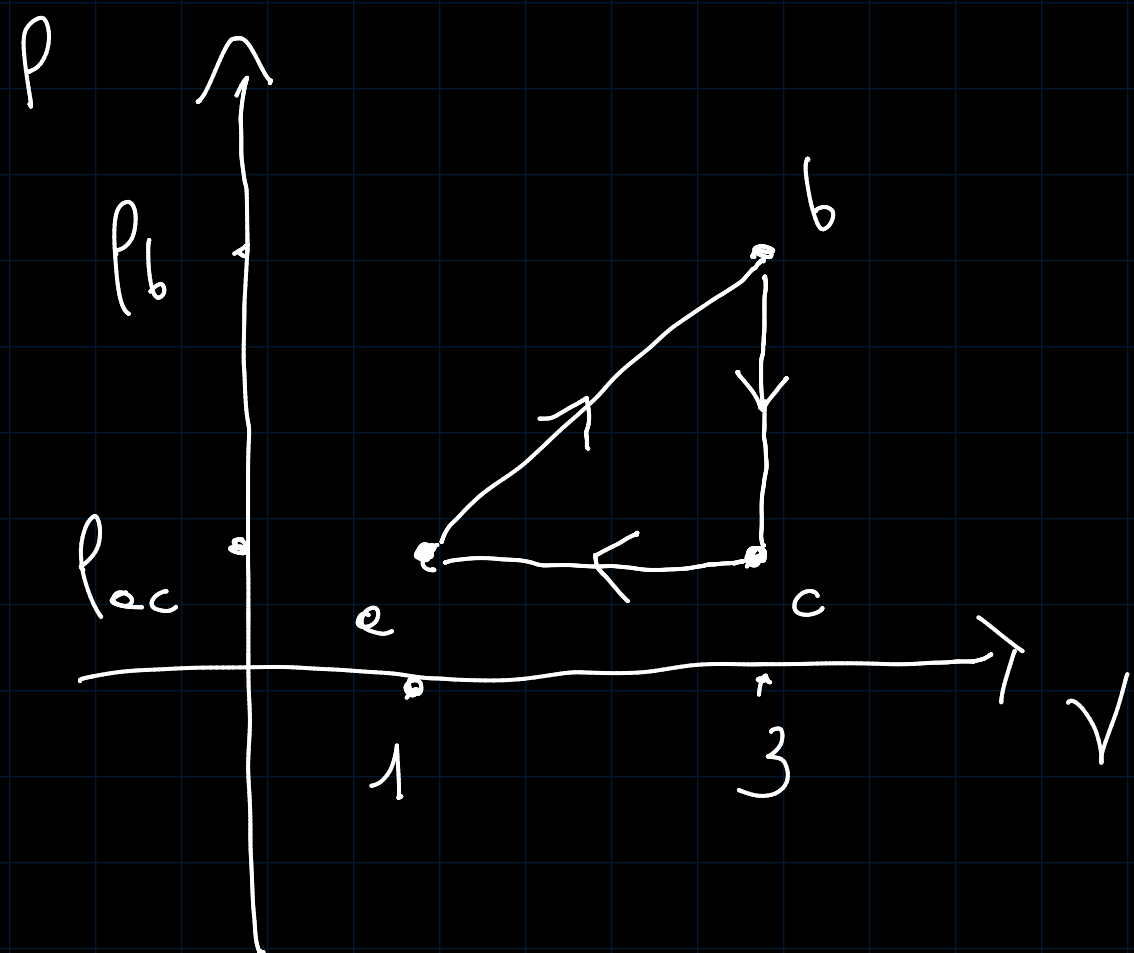
$$\begin{aligned} L &= p \Delta V = p_f (V_i - V_f) = p_f \left(V_i - \frac{p_i}{p_f} V_i \right) = p_f V_i - p_i V_i \\ &= V_i (p_f - p_i) = (0,140) \cdot (101,3 \cdot 10^3 - 204 \cdot 10^3) \\ &= -143765 \end{aligned}$$

$$L_{\text{tot}} = 199935 \text{ J} - 143765 \text{ J} = 56170 \text{ J}$$

13) $p_c = 2,5 \text{ kPa} = 2,5 \cdot 10^3 \text{ Pa}$

$p_b = 7,5 \text{ kPa} = 7,5 \cdot 10^3 \text{ Pa}$

$T_c = 200^\circ \text{ K}$



$$pV = nRT$$

a) $n = \frac{p_e V_e}{R T_e} = \frac{2,5 \cdot 10^3 \cdot 1}{8,31 \cdot 200} = 1,5 \text{ mol}$

b) $T_b = \frac{p_b V_b}{nR} = \frac{7,5 \cdot 10^3 \cdot 3}{1,5 \cdot 8,31} = 1805^\circ \text{ K}$

c) $T_c = \frac{p_c V_c}{nR} = \frac{2,5 \cdot 10^3 \cdot 3}{1,5 \cdot 8,31} = 601,7^\circ \text{ K}$

d) Q: $\Delta E_{int} = 0 \Rightarrow Q = L$

$$L = \frac{1}{2} (2 \text{ m}^3) (5 \cdot 10^3 \text{ Pa}) = 5 \cdot 10^3 \text{ J}$$

42) $n = 1 \text{ mol}$
 $T = 273^\circ \text{K}$ / $E_{int}?$

$$E_{int} = \frac{3}{2} n R T = \frac{3}{2} (1 \text{ mol}) \left(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (273^\circ \text{K}) = 3403 \text{ J}$$

43) $n = 3 \text{ mol}$
 $\Delta T = 20^\circ \text{C} = 313,15^\circ \text{K}$
 $p \text{ è costante}$ / $Q?$
 $\Delta E_{int}?$
 $L?$

$$L = n R \Delta T = (3 \text{ mol}) \left(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) (313,15^\circ \text{K}) = 7807 \text{ J}$$

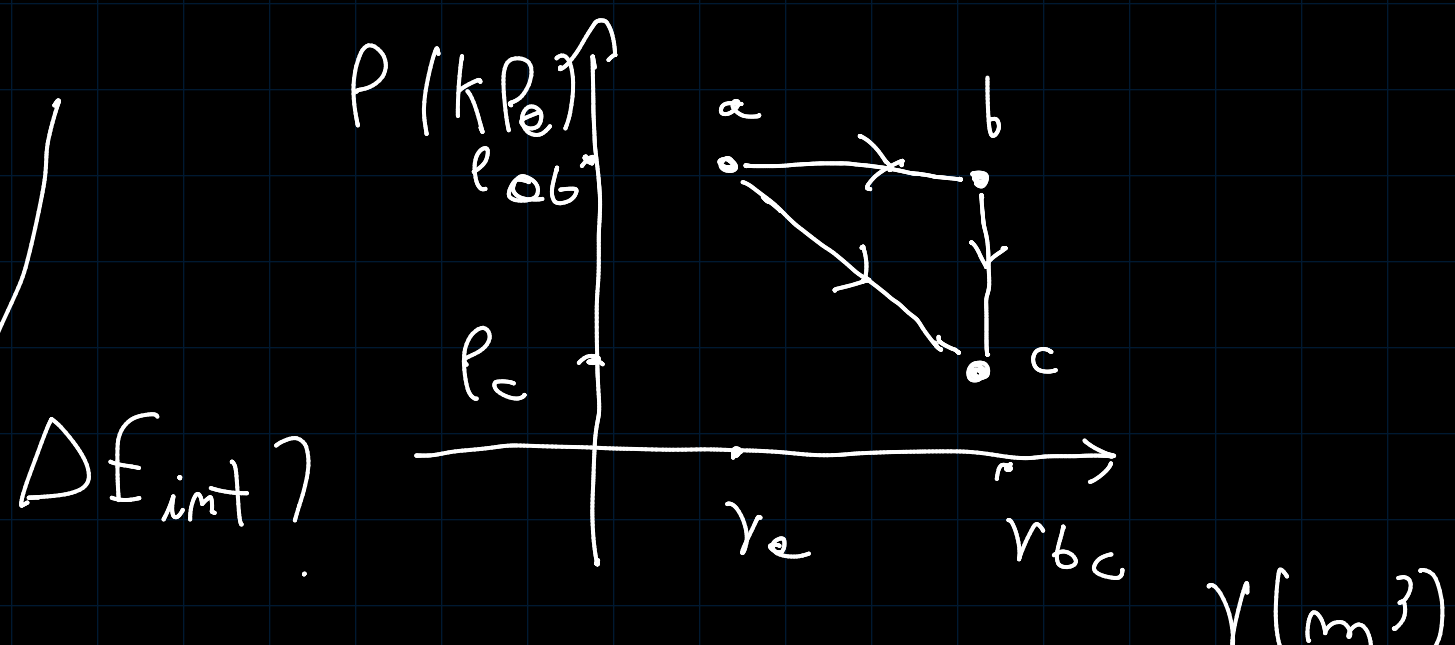
$$Q = n c_p \Delta T = n \frac{7}{2} R \cdot \Delta T = (3 \text{ mol}) \left(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}} \right) \frac{7}{2} (313,15^\circ \text{K}) = 27324 \text{ J}$$

$$\Delta E_{int} = Q - L = 27324 \text{ J} - 7807 \text{ J} = 19517 \text{ J}$$

44) $n = 1 \text{ mol}$

$$p_{ab} = 5 \text{ kPa} = 5 \cdot 10^3 \text{ Pa}$$

$$p_c = 1 \text{ kPa} = 1 \cdot 10^3 \text{ Pa}$$



$$p_c = 2 \text{ kPa} = 2 \cdot 10^3 \text{ Pa} \quad / \quad Q?$$

$$V_c = 4 \text{ m}^3 \quad / \quad Q_{abc}?$$

$$V_a = 2 \text{ m}^3$$

$$p_a V_a = n R T_a \quad \Rightarrow \text{SooP}$$

$$T_a = \frac{p_a V_a}{n R} = \frac{3 \cdot 10^3 \cdot 2}{8,31} = 1203^\circ \text{K}$$

$$T_c = \frac{p_c V_c}{n R} = \frac{2 \cdot 10^3 \cdot 4}{8,31} = 962,7^\circ \text{K}$$

$$\Delta E_{\text{int}} = n C_V \Delta T = n \frac{5}{2} R \left(\frac{p_c V_c}{n R} - \frac{p_a V_a}{n R} \right) = -5000 \text{ J}$$

$$L = \frac{p_a + p_c}{2} (V_c - V_a) = 7000 \text{ J}$$

$$Q = \Delta E_{\text{int}} + L = -5000 \text{ J} + 7000 \text{ J} = 2000 \text{ J}$$

$$Q_{abc} = Q_{ab} + Q_{bc} \quad / \quad p \text{ const} \quad / \quad V \text{ const}$$

$$= p_a \Delta V + \Delta E_{\text{int}} = n \frac{7}{2} R \Delta T + \Delta E_{\text{int}}$$

$$= 10000 \text{ J} - \Delta E_{\text{int}}$$

$$= 10000 \text{ J} - 5000 \text{ J} = 5000 \text{ J}$$

46) $n = 2 \text{ mol}$ $/$ $L?$ $\Delta E_{\text{int}}?$
 $\Delta T = 15^\circ \text{K}$ $/$ $Q?$ $\Delta K?$

p costante

$$pV = nRT \quad \Delta E_{int} = Q - L$$

$$L = nR\Delta T = (2 \text{ mol}) \left(8,31 \frac{\text{J}}{\text{mol}^\circ\text{K}} \right) (15^\circ\text{K}) = 249,3 \text{ J}$$

$$Q = nC_p\Delta T = (2 \text{ mol}) \frac{5}{2} R (15^\circ\text{K}) = (2 \text{ mol}) \left(\frac{5}{2} 8,31 \frac{\text{J}}{\text{mol}^\circ\text{K}} \right) (15^\circ\text{K}) \\ = 623,25 \text{ J}$$

$$\Delta E_{int} = Q - L = (623,25 - 249,3) \text{ J} = 374 \text{ J}$$

$$\Delta K = \frac{\Delta E_{int}}{N} = \frac{\Delta E_{int}}{n N_A} = \frac{374 \text{ J}}{(2 \text{ mol}) (6,02 \cdot 10^{23} \text{ mol}^{-1})} = 3,11 \cdot 10^{-22} \text{ J}$$

(47) $n = 2 \text{ mol}$ / $L?$ $\Delta K?$
 $\Delta T = 15^\circ\text{K}$ / $Q?$
 V costante / $\Delta E_{int}?$

$$pV = nRT \quad \Delta E_{int} = Q - L$$

$$\text{Se } V \text{ \u00e9 costante} \Rightarrow L = 0$$

$$Q = nC_v\Delta T = n \frac{3}{2} R \Delta T = (2 \text{ mol}) \frac{3}{2} \left(8,31 \frac{\text{J}}{\text{mol}^\circ\text{K}} \right) (15^\circ\text{K}) = \\ = 374 \text{ J}$$

$$\Delta E_{int} = Q - L = Q = 374 \text{ J}$$

$$\Delta K = \frac{\Delta E_{int}}{N} = \frac{\Delta E_{int}}{n N_A} = \frac{374 \text{ J}}{(2 \text{ mol}) (6,02 \cdot 10^{23} \text{ mol}^{-1})} = 3,11 \cdot 10^{-22} \text{ J}$$

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$$Q = 20,9 \text{ J}$$

$$V_i = 50 \text{ cm}^3 = 5 \cdot 10^{-5} \text{ m}^3$$

$$V_f = 100 \text{ cm}^3 = 1 \cdot 10^{-4} \text{ m}^3$$

$$p = 1 \text{ atm} = 1 \cdot 10^5 \text{ Pa (costante)}$$

$$\Delta E_{\text{int}}?$$

$$\text{Se } n = 2 \cdot 10^{-3} \text{ mol}$$

$$\text{trovare } C_p, C_v?$$

$$L = p \Delta V = (1 \cdot 10^5 \text{ Pa}) (1 \cdot 10^{-4} - 5 \cdot 10^{-5}) \text{ m}^3 = 5 \text{ J}$$

$$\Delta E_{\text{int}} = Q - L = 20,9 \text{ J} - 5 \text{ J} = 15,9 \text{ J}$$

$$Q = n C_p \Delta T \Rightarrow C_p = \frac{Q}{n \Delta T}$$

$$\Delta E_{\text{int}} = n C_v \Delta T$$

$$L = n R \Delta T \Rightarrow \Delta T = \frac{L}{n R} = \frac{5 \text{ J}}{(2 \cdot 10^{-3} \text{ mol}) (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}})} = 300,8^\circ \text{K}$$

$$C_p = \frac{Q}{n \Delta T} = \frac{20,9 \text{ J}}{(2 \cdot 10^{-3} \text{ mol}) (300,8 \text{ K})} = 34,8 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

$$C_v = C_p - R = (34,8 - 8,31) \frac{\text{J}}{\text{mol} \cdot \text{K}} = 26,5 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

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$$n_1 = 2,40 \text{ mol}$$

$$C_{v1} = 12 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

$$n_2 = 1,5 \text{ mol}$$

$$C_{v2} = 12,8 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

$$n_3 = 3,7 \text{ mol}$$

$$C_v?$$

$$c_{r3} = 20 \frac{\text{J}}{\text{mol} \cdot ^\circ\text{K}}$$

$$C_V = \frac{(n_1 c_{r1}) + (n_2 c_{r2}) + (n_3 c_{r3})}{n_1 + n_2 + n_3} = 15.8 \frac{\text{J}}{\text{mol} \cdot ^\circ\text{K}}$$

30) $Q = 70 \text{ J}$
 gas biatómico
 p constante / $\Delta E_{\text{int}}?$

$$Q = n c_p \Delta T = n \frac{7}{2} R \Delta T \quad \Delta E_{\text{int}} = Q - L$$

$$\Delta T = \frac{Q}{n \frac{7}{2} R} \quad L = n R \Delta T = n R \frac{Q}{n \frac{7}{2} R} = \frac{2Q}{7} = 20 \text{ J}$$

$$\Delta E_{\text{int}} = Q - L = 50 \text{ J}$$

51) $n = 1 \text{ mol}$
 p constante / $Q?$
 $T_i = 0^\circ\text{C} = 273,15^\circ\text{K}$

$$Q = n c_p \Delta T \quad \Delta E_{\text{int}} = n c_v \Delta T \quad L = n R \Delta T$$

$$Q = n \frac{7}{2} R \Delta T = n \frac{7}{2} R (2T_i - T_i) =$$

$$= n \frac{7}{2} R T_i = 7944,6 \text{ J}$$

$$\frac{p_i V_i = n R T_i}{p_f 2V_i n R T_f} \Rightarrow \frac{1}{2} = \frac{T_i}{T_f}$$

$$\Rightarrow 2 = \frac{T_f}{T_i} \Rightarrow T_f = 2T_i$$

52) $M = 12 \text{ u} = 0,012 \text{ kg}$ /

p costante

$Q?$

$$T_i = 25^\circ\text{C} = 298,15^\circ\text{K}$$

$$T_f = 125^\circ\text{C} = 398,15^\circ\text{K}$$

$$n = \frac{M_{\text{compiome}}}{m_{\text{mol}} \cdot N_A} = \frac{0,012 \text{ kg}}{32 \cdot 10^{-3} \frac{\text{kg}}{\text{mol}}} = 0,375$$

$$Q = n c_p \Delta T = n \frac{7}{2} R \Delta T = (0,375 \text{ mol}) \frac{7}{2} (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (100^\circ\text{K}) = 1090,7 \text{ J}$$

33) $n = 4 \text{ mol}$ / $Q?$

$\Delta T = 60^\circ\text{K}$ / $\Delta E_{\text{int}}?$

p costante / $L?$

$$Q = n c_p \Delta T = n \frac{7}{2} R \Delta T = (4 \text{ mol}) \frac{7}{2} (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (60^\circ\text{K}) = 6980,4 \text{ J}$$

$$\Delta E_{\text{int}} = n c_v \Delta T = n \frac{5}{2} R \Delta T = (4 \text{ mol}) \frac{5}{2} (8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}}) (60^\circ\text{K}) = 4986 \text{ J}$$

$$L = Q - \Delta E_{\text{int}} = 6980,4 \text{ J} - 4986 \text{ J} = 1994,4 \text{ J}$$

57) $V_i = 200 \text{ L} = 2 \cdot 10^{-1} \text{ m}^3$

$V_f = 74,3 \text{ L} = 0,0743 \text{ m}^3$

$p_i = 10^5 \text{ Pa}$

$T_i = 300^\circ\text{K}$

mono, bi o poli?

$n?$

$T_f?$

$$p_f = 4 \text{ atm} = 4 \cdot 10^5 \text{ Pa}$$

$$pV = nR\bar{T} \Rightarrow n = \frac{pV_i}{R\bar{T}_i} = \frac{(10^5 \text{ Pa})(2 \cdot 10^{-1} \text{ m}^3)}{(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}})(300 \text{ K})} = \frac{20000 \text{ J}}{2493 \frac{\text{J}}{\text{mol}}} = 8,02 \text{ mol}$$

$$\bar{T}_f = \frac{p_f V_f}{nR} = \frac{(4 \cdot 10^5 \text{ Pa})(0,0743 \text{ m}^3)}{(8,02 \text{ mol})(8,31 \frac{\text{J}}{\text{mol} \cdot \text{K}})} = \frac{29720 \text{ J}}{66,7 \frac{\text{J}}{\text{K}}} = 445,6 \text{ K}$$

$$p_i V_i^\gamma = p_f V_f^\gamma$$

$$\frac{p_i}{p_f} = \left(\frac{V_f}{V_i} \right)^\gamma$$

$$\ln\left(\frac{p_i}{p_f}\right) = \gamma \ln\left(\frac{V_f}{V_i}\right)$$

$$\gamma = \frac{\ln\left(\frac{p_i}{p_f}\right)}{\ln\left(\frac{V_f}{V_i}\right)} = \frac{\ln(0,25)}{\ln(0,37)} = \frac{-1,4}{-1} = 1,4 = \frac{7}{5} \text{ (biatomic)}$$

33) $V_i = 4,3 \text{ L} = 0,043 \text{ m}^3$

$$p = 1,2 \text{ atm} = 1,2 \cdot 10^5 \text{ Pa}$$

$$\bar{T}_i = 310 \text{ K}$$

$$V_f = 0,76 \text{ L} = 0,00076 \text{ m}^3$$

$$\gamma = 1,4 = \frac{7}{5}$$

p_f ?

$\bar{T}_f$?

$$p_i v_i^\gamma = p_f v_f^\gamma$$

$$p_f = \frac{p_i v_i^\gamma}{v_f^\gamma} = \frac{1,2 \cdot 10^5 \text{ Pa} (0,043 \text{ m}^3)^{1,4}}{(0,00076 \text{ m}^3)^{1,4}} = 3,4 \cdot 10^7 \text{ Pa}$$

$$p_i v_i = n R T_i \Rightarrow n = \frac{p_i v_i}{R T_i} = \frac{(1,2 \cdot 10^5 \text{ Pa})(0,043 \text{ m}^3)}{8,31 \cdot 310 \text{ K}} = 2 \text{ mol}$$

$$p_f v_f = n R T_f$$

$$T_f = \frac{p_f v_f}{n R} = \frac{(3,4 \cdot 10^7 \text{ Pa})(0,00076 \text{ m}^3)}{2 \cdot 8,31} = 1554,8 \text{ K}$$

58) $p_i = 5 \text{ atm} = 5 \cdot 10^5 \text{ Pa}$

$$p_f = 1 \text{ atm} = 10^5 \text{ Pa}$$

$$\gamma = \frac{4}{3}$$

$$T_i = 5^\circ \text{C} = 278,15 \text{ K}$$

$$T_f ?$$

$$p V = n R T \Rightarrow V = \frac{T}{p} n R$$

$$p_i \left(\frac{T_i}{p_i} \right)^\gamma = p_f \left(\frac{T_f}{p_f} \right)^\gamma$$

$$p_i^{1-\gamma} T_i^\gamma = p_f^{1-\gamma} T_f^\gamma$$

$$(1-\gamma) \ln(p_i) + \gamma \ln(T_i) = (1-\gamma) \ln(p_f) + \gamma \ln(T_f)$$

$$\ln(T_f) = \frac{(1-\gamma) \ln(p_i) + \gamma \ln(T_i)}{(1-\gamma) \ln(p_f) + \gamma} = 6,4$$

$$T_f = 2^{6.4} = 602^\circ \text{K}$$

59. •• La Figura 19.26 mostra due percorsi che possono essere seguiti da un gas da un punto iniziale i a un punto finale f . Il percorso 1 è formato da un'espansione isoterma (il lavoro è 50 J in valore assoluto), un'espansione adiabatica (il lavoro è 40 J in valore assoluto), una compressione isoterma (il lavoro è 30 J in valore assoluto) e infine una compressione adiabatica (il lavoro è 25 J in valore assoluto). Qual è la variazione dell'energia interna del gas se questo va dal punto i al punto f seguendo il percorso 2?

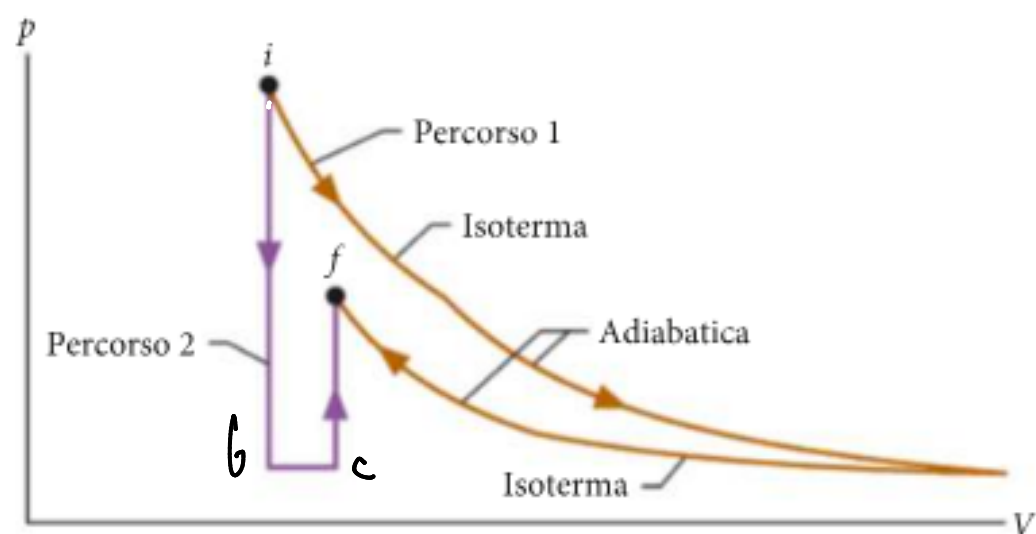


Figura 19.26 Problema 59.

ΔE_{int} non dipende dal percorso $\Rightarrow \Delta E_{\text{int},1} = \Delta E_{\text{int},2}$

$$\Delta E_{\text{int},1} = \Delta E_{\text{int},\text{iso}} + \Delta E_{\text{int},\text{adie}} + \Delta E_{\text{int},\text{iso}} + \Delta E_{\text{int},\text{adie}}$$

In un'espansione isoterma $\Delta E_{\text{int}} = 0$ poiché ΔE_{int}

dipende dalla temperatura e in un'isoterma $\Delta T = 0$

$$\begin{aligned} \Delta E_{\text{int},1} &= \Delta E_{\text{int},\text{adie}} + \Delta E_{\text{int},\text{adie}} = -L_1 - L_2 = -40\text{J} - (-25\text{J}) \\ &= -15\text{J} = \Delta E_{\text{int},2} \end{aligned}$$